Sequence 13

SYNTHESIS PROBLEMS, REASONING AND FLUENCY

30 March–9 April 2027

I Implication, converse and counterexample

Tools to use in context

"If P, then Q" is an implication $P \Rightarrow Q$. Its converse is $Q \Rightarrow P$; it is not automatically true. An equivalence $P \iff Q$ requires both implications. A counterexample is enough to refute a universal statement; a few examples are not enough to prove it. The contrapositive not $Q \Rightarrow$ not P is equivalent to $P \Rightarrow Q$.

Example 13.1

For an integer n: "If n is a multiple of 4, then n is even." Prove the implication, write its converse and its contrapositive, then say which are true.

Solution:

If $n = 4k$, then $n = 2(2k)$ is even. Converse: "If n is even, then n is a multiple of 4"; it is false with $n = 2$. Contrapositive: "If n is odd, then n is not a multiple of 4"; it is true and equivalent to the original implication.

Example 13.2

For $x \in \mathbb{R}$, study "$x = 3 \Rightarrow x^2 = 9$" and its converse. Give a correct equivalence.

Solution:

The implication is true. The converse is false with $x = -3$. The correct equivalence is $x^2 = 9 \iff x = 3$ or $x = -3$. With the extra hypothesis $x \geq 0$, one can write $x^2 = 9 \iff x = 3$.

Teacher notes

These notions are made explicit from the problems; they are not a standalone chapter of formal logic to memorise without context.

II Negation, case analysis and proof

Example 13.3

Negate correctly: "$x > 2$"; "$x \in [1; 4]$"; "All students play sport"; "A and B both occur".

Solution:

> "$x \leq 2$"; "$x < 1$ or $x > 4$"; "At least one student does not play sport"; "A does not occur or B does not occur". The "or" is inclusive.

Example 13.4

Show by case analysis that $n(n+1)$ is even for every integer n. Then show that $n^2 + n + 1$ is odd.

Solution:

> If n is even, $n(n+1)$ has an even factor. If n is odd, $n+1$ is even; the product is again even. The two cases cover all integers. Since $n^2 + n + 1 = n(n+1) + 1$, it is an odd integer.

Example 13.5

A student writes "n^2 even implies n even, because I checked it for $n = 2, 4, 6$". Replace this justification with a proof.

Solution:

> If n were odd, the result proved in sequence 1 would give n^2 odd, a contradiction. This is a contrapositive argument, or presented here as a proof by contradiction. Three examples do not cover all integers.

III Combining algebra, functions and percentages

Example 13.6: Original situation

Two fictional rental rates are offered for x hours, with $x \in [0;8]$: $A(x) = 3000+2000x$ Ar and $B(x) = 3500x$ Ar.

1. Calculate the costs for 1 h and 4 h.
2. Find where the rates are equal, then the durations for which A is cheaper.
3. A 10 % discount is granted on the whole of rate A. Give the new rate A_r and the new comparison threshold.

Solution:

1. At 1 h: 5000 Ar versus 3500 Ar. At 4 h: 11000 Ar versus 14000 Ar. 2. $3000+2000x = 3500x \iff x = 2$. $A < B$ for $x > 2$, so on $]2;8]$. 3. $A_r(x) = 0.9(3000+2000x) = 2700+1800x$. Equality with B gives $1700x = 2700$, that is $x = 27/17 \approx 1.59$ h. $A_r < B$ on $]27/17;8]$. The discount also applies to the fixed charge.

```
def tarif_moins_cher(x):
    a = 3000 + 2000*x
    b = 3500*x
    if a < b:
        return "A"
    elif b < a:
        return "B"
    else:
        return "egalite"
```

Example 13.7

Predict the results for $x = 1, 2, 4$. Modify the line that computes a to take the discount into account.

Solution:

Results: `"B"`, `"egalite"`, `"A"`. Replace with `a = 0.9*(3000 + 2000*x)`. The domain $[0;8]$ is assumed to be respected in the calls.